Teacher Wu Kuan | See It at a Glance #10 — $(a + b)^2$ Is Not $a^2 + b^2$

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1. The trap question

Is $(a + b)^2 = a^2 + b^2$?

2. The method

- Test $a = b = 1$: 4 vs 2 → not equal.
- Square of side $a + b$ splits into a^2, ab, ab, b^2.
- $(a + b)^2 = a^2 + 2ab + b^2$ — the missing part is $2ab$.

3. Parent check method

Have your child test any formula with small numbers before trusting it.

Practice

1. Expand $(x + 3)^2$.
2. Expand $(2x + 1)^2$.
3. Use $(20 + 1)^2$ to find 21^2.
4. Simplify $(a + b)^2 - (a^2 + b^2)$.
5. $a + b = 5$ and $ab = 6$. Find $a^2 + b^2$.

Answers

1. $x^2 + 6x + 9$
2. $4x^2 + 4x + 1$
3. 441
4. $2ab$
5. 13

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